Fine-Tuning a Language Model for Bird Conservation Report Drafting

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Abstract

Preparing a conservation report from incomplete biodiversity information requires more than a fluent summary: the report must preserve the distinction between available evidence and unanswered questions. We investigated whether parameter-efficient fine-tuning could help a language model perform this task within EcoPath, a geospatial bird-intelligence platform. An archived QLoRA experiment adapted Qwen2.5-7B-Instruct on 85 synthetic examples with taxonomy-only inputs. On 15 held-out species, valid JSON and required-key presence were observed in 15/15 adapted responses and 0/15 base responses; the original prompt did not explicitly require JSON. We then compared a newly trained adapter with a strongly prompted unadapted model on 60 additional held-out species under three evidence variants, using identical prompts and schema constraints. End-to-end field success was 100.0% for QLoRA and 72.1% for the unadapted comparator (paired difference +27.9 percentage points; 95% species bootstrap interval +27.2 to +29.0). Available-value agreement, abstention and source associations were assessed separately. This single-seed comparison evaluates agreement with reviewed structured assertions, rather than independent ecological correctness. EcoPath retains a bounded report-drafting role while the measured results qualify the incremental contribution of adaptation.

Keywords: ecological AI; biodiversity data platform; QLoRA; bird conservation; report drafting; structured generation; missing evidence.

1 Introduction

Conservation reporting depends on assembling information that differs in its source, coverage, and relevance to the question being asked. A species may have an identifiable taxonomic record and a traceable occurrence source while still lacking the population, habitat, or threat evidence needed for an assessment. For bird-related information systems, the practical challenge is therefore not simply to retrieve records, but to communicate what those records support and where the evidence remains incomplete.

Citizen-science initiatives and open repositories have expanded opportunities to work with biodiversity information. The eBird framework illustrates how distributed observations can support avian research [1], while research on GBIF demonstrates that spatial bias can affect the interpretation of aggregated records [2]. These resources are valuable inputs to conservation workflows, but their availability should not be confused with sufficient evidence for every field in a report. A reporting system must preserve the distinction between observed information, unavailable information, and conclusions requiring further investigation.

Large language models (LLMs) offer a way to turn heterogeneous information into readable drafts. Castro et al. [9] demonstrated their potential for extracting and structuring ecological information from text, although performance differed across tasks and models. The same fluency that makes these systems useful can also conceal unsupported content [7]. In a conservation report, a plausible description of threats may

appear informative even when those threats were absent from the input. Useful drafting therefore requires both a consistent report structure and a response to uncertainty that readers can inspect.

There are several ways to address these requirements. Retrieval-augmented generation supplies external information to the generator [3], while output specifications define how that information should be presented. Parameter-efficient adaptation offers a further possibility: learning a reporting pattern from examples without updating the full model [4,6]. However, producing the requested structure and faithfully using evidence are different achievements. Format restrictions can also affect task performance [10]. An evaluation of report drafting must consequently distinguish successful delivery of an output format from the quality of the ecological content within it.

We examined this problem within EcoPath, a platform that combines bird species exploration, geospatial visualisation, and AI-assisted interaction. Its report-drafting module is intended to organise selected information for human review. We used QLoRA to adapt Qwen2.5-7B-Instruct [11] on examples in which major ecological fields were missing and compared its responses with those of the unadapted model. This setting allowed us to examine whether a small training dataset could establish a consistent report format and encourage explicit acknowledgement of missing information.

The study addressed two questions: (i) how does QLoRA adaptation change structural compliance on held-out species under the same reporting prompts, and (ii) what differences in missing-information handling are apparent in automated ratings and illustrative outputs? We expected the adapted model to follow the report schema more consistently. We treated the second question as exploratory because the original inputs did not vary the principal ecological fields. A subsequent matched comparison on 60 held-out species (Appendix B) tested supplied-evidence use under identical prompting and schema constraints.

2 Related Work

The reporting problem connects research on biodiversity information with work on language-model grounding, structured generation, and efficient adaptation. The literature provides reasons to investigate each component, but also makes clear why success on one component cannot substitute for evaluation of the others.

2.1 Biodiversity Information and Ecological AI

Sullivan et al. [1] describe how eBird supports the collection and reuse of bird observations through a shared information system. Beck et al. [2] examine a complementary issue: spatial bias in aggregated occurrence records can affect downstream analyses. For report drafting, these studies motivate retaining source context and avoiding conclusions that exceed the observations. Tuia et al. [5] further emphasise the role of ecological knowledge and interdisciplinary collaboration in machine-learning applications for conservation. We therefore place EcoPath within an evidence-inspection workflow in which automated drafting supports, rather than determines, conservation interpretation.

2.2 Language Models for Ecological Information Processing

Castro et al. [9] evaluated prompt-based LLMs for identifying, extracting, geocoding, and structuring species-distribution information. Their work demonstrates that language models can assist ecological data processing and that their performance depends on the task. This motivates a separate assessment of report structure and content in EcoPath. It also establishes prompting as an important comparator when determining whether additional model training is worthwhile.

Ecological knowledge benchmarks show why evidence organisation should be separated from judgement. Dorm et al. [12] found differences in LLM performance across species presence, range mapping, threat classification, and related tasks. Uryu [13], in a preprint examining IUCN-related species information, likewise reported differing performance on taxonomic and conservation-status questions. These studies do

not evaluate the same task as EcoPath, but they motivate its restricted scope: organising the supplied material without treating the draft as a formal conservation assessment.

2.3 Grounding and Structured Generation

Lewis et al. [3] introduced retrieval-augmented generation as a means of conditioning language generation on external information. For a report generator, such access is useful only if the output remains faithful to the retrieved material. Structure adds a separate requirement. Tam et al. [10] found that format restrictions can affect performance on reasoning tasks, showing that predictable output and successful task execution need independent measurement. EcoPath accordingly treats JSON validity as an integration measure and evidence support as a distinct question.

2.4 Parameter-Efficient Adaptation and Evaluation

LoRA reduces the number of parameters updated during adaptation by learning low-rank changes to frozen pretrained weights [4]. QLoRA extends this approach to a quantised backbone, reducing the memory needed for training [6]. These properties make adaptation relevant to small application-development teams, although they do not establish a quality or cost advantage for a particular workflow. We used the instruction-tuned Qwen2.5 family [11] to examine adaptation in the specific context of sparse-input report drafting.

Evaluating the resulting text introduces a further challenge. Zheng et al. [8] showed that LLM-based judges can approximate human preferences while exhibiting position, verbosity, and self-enhancement biases. Automated scores can therefore provide an exploratory comparison, but do not independently verify ecological statements. The present study combines deterministic format checks with supplementary judging and qualitative examples, while reserving claims about evidence fidelity for more direct evaluation.

3 Methodology

We implemented the drafting task within EcoPath and evaluated the same instruction-tuned model with and without a QLoRA adapter. Sections 3-5 describe the archived synthetic-data experiment and its 15-species comparison. Appendix B reports a separate sourced dataset, new training and a matched 60-species evaluation addressing the same drafting purpose.

3.1 System Architecture

EcoPath separates user interaction, application orchestration, persistent data, external sources, and AI services (Figure 1; Table 1). A Next.js frontend provides species search, profiles, image-assisted identification, and map-based exploration, with MapLibre GL JS supporting visualisation. FastAPI handles requests and coordinates species-resolution and service workflows. Supabase and PostgreSQL provide persistent storage, while PostGIS supports spatial queries.

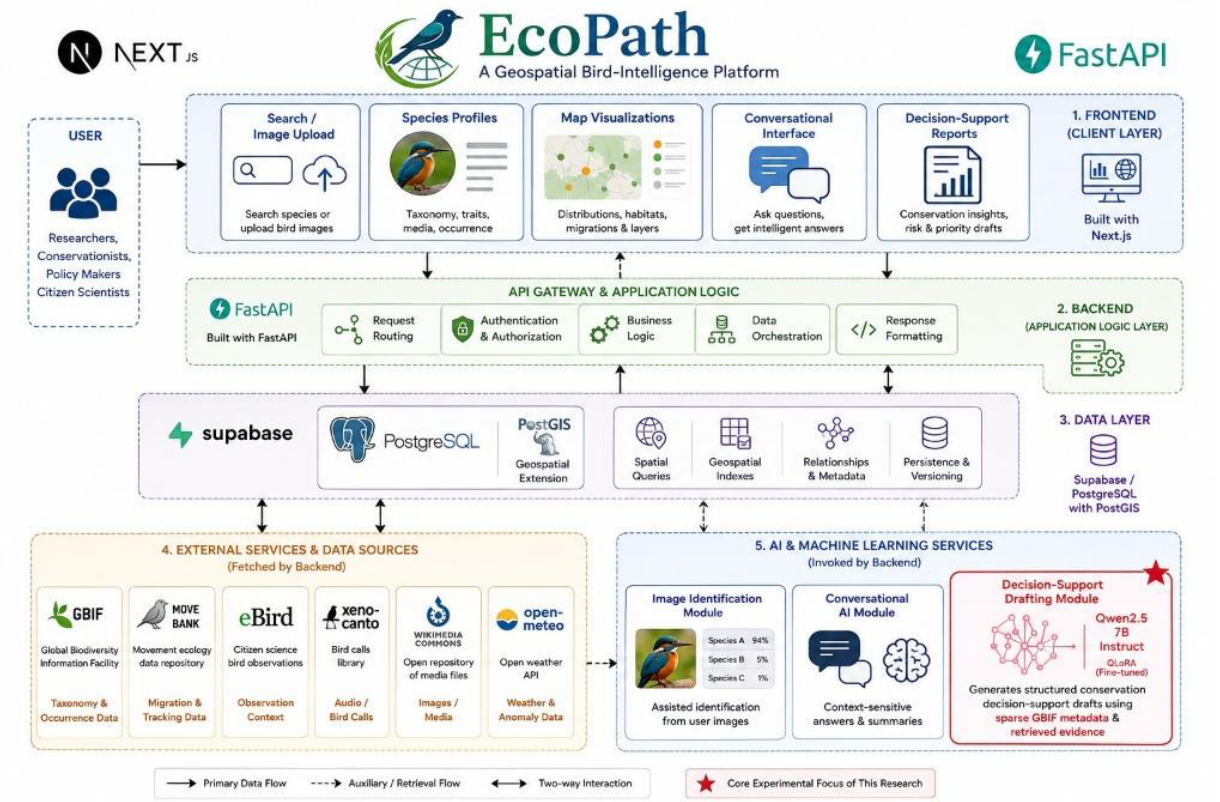


Figure 1. EcoPath architecture. The report-drafting module is the experimental focus; the remaining platform components provide the application context.

The platform connects to GBIF, Movebank, eBird, Xeno-canto, Wikimedia Commons, and Open-Meteo for different biodiversity, movement, media, and weather functions. Its AI services support identification, conversation, and report drafting. Our evaluation concerned the drafting module alone. Although several sources are accessible through the platform, the experimental inputs contained only the GBIF-linked metadata described in Section 3.3.

Table 1. EcoPath architecture and the scope of the report-generation experiment.

Layer	Responsibilities and technologies	Experimental scope
Presentation	Search, profiles, maps, report display; Next.js and MapLibre GL JS	Application context
Application services	Routing, species resolution, orchestration; FastAPI	Application context
Persistent data	Records and spatial queries; Supabase, PostgreSQL, PostGIS	Application context
External services	Biodiversity, movement, media, and weather sources	Only supplied GBIF-linked metadata evaluated
AI services	Identification, conversation, and QLoRA report drafting	Report drafting evaluated

3.2 Evidence-Bounded Decision-Support Pipeline

The workflow consisted of preparing species-linked targets, dividing the dataset into training and held-out subsets, adapting the model, and generating paired reports (Figure 2). Each model condition received the same held-out species payload and reporting prompt. We assessed parseability and required-key presence, then obtained supplementary automated ratings. The archived prompt did not explicitly require JSON or supply a schema; Appendix B adds a matched, explicitly constrained comparison.



Figure 2. Dataset preparation and QLoRA adaptation. The unadapted model and the resulting adapter were compared using identical inputs for 15 held-out species.

The intended evidence boundary was straightforward: organise the supplied information, retain the source reference, and acknowledge unavailable fields. The pipeline did not infer an official IUCN category or verify the draft against an independent ecological assessment. Human review was the intended use context, rather than an evaluated intervention in this experiment.

3.3 Dataset Generation and Evidence Boundary

We constructed the EcoPath Gemini-100 supervised fine-tuning dataset, comprising 100 synthetic examples for 100 bird species. Each example contained a structured species payload, retrieval queries, evidence records, and a teacher-generated report. Species identity and taxonomic metadata were resolved through GBIF. A GBIF URL provided the traceable source reference, while verified IUCN status, population trend, population estimate, habitat, and range fields were absent.

Target reports identified many evidence gaps, exposing the model to a consistent structured drafting pattern. However, the synthetic targets were not independently verified conservation references: source auditing also identified heuristic risk language unsupported by the taxonomy-only inputs. Because the principal ecological fields were absent throughout, this dataset could not test whether the model would selectively report supported values when richer evidence became available.

Using a deterministic split with seed 42, we assigned 85 species examples to training and 15 to held-out evaluation. There was no separate validation partition or recorded validation-based checkpoint selection. Species were disjoint, but Larus and Turdus occurred in both splits. The evaluation species are listed in Appendix A. Although EcoPath has an India-focused motivation, this geographically broader held-out list was not designed to estimate performance specifically on Indian birds.

3.4 Fine-Tuning Configuration

We loaded Qwen/Qwen2.5-7B-Instruct [11] in 4-bit quantised form and trained a QLoRA adapter [6]. Updates targeted the q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, and down_proj layers. The adapter used rank 16, alpha 32, and dropout 0.05. Training proceeded for three epochs with a learning rate of 2×10^{-4} and batch size 1 with gradient accumulation, giving 33 optimisation steps (Table 2).

Table 2. Reported fine-tuning configuration.

Parameter	Value
Base model	Qwen/Qwen2.5-7B-Instruct
Quantisation	4-bit backbone
Training / held-out examples	85 / 15; split seed 42
LoRA rank / alpha / dropout	16 / 32 / 0.05
Target layers	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Learning rate	2×10^{-4}
Batch size	1, with gradient accumulation
Duration	3 epochs; 33 optimisation steps
Recorded loss values	1.437 at step 10; 0.376 at step 30
Independent training runs	One reported
Validation set	None reported

We conducted one adaptation run. Training loss was recorded during optimisation, and the resulting adapter was used for the held-out comparison. No validation-based checkpoint comparison or repeated-seed analysis was conducted. The implications of this design and incomplete implementation reporting are addressed in Section 5.5.

3.5 Evaluation Protocol

We generated reports for each of the 15 held-out species in two conditions: the unadapted Qwen2.5-7B-Instruct model and the same model with the EcoPath adapter. The conditions used the same prompts, evidence payloads, and generation settings. This paired design held the input constant while examining the difference associated with the adapter under the tested configuration.

We assessed JSON validity after removing optional Markdown fences, then checked the presence of six component keys: species, current conservation status, key threats, survivability assessment, mitigation actions, and sources. The historical check accepted flat or nested layouts and did not validate field types or a full JSON schema. These measures describe parseability and key presence; they do not verify factual support. Full schema validation is used in Appendix B.

A Gemini-family LLM judge assigned five-point ratings for format, grounding, completeness and usefulness to all 30 responses. Its context was limited to the first 600 characters of the user prompt, generally omitting the source-result section. These historical ratings therefore do not establish fidelity to the full supplied evidence. No blinded human validation or teacher-judge independence was established.

We summarised structural outcomes as counts and proportions, report length as the mean number of words, and rubric outcomes as descriptive means. For each binary structural proportion, we calculated a two-sided 95% exact binomial interval. These intervals describe the observed proportions under a binomial assumption; they are not confidence intervals for the paired treatment difference. We did not estimate an unsupported-claim rate, field-level abstention accuracy, or citation-support rate. Qualitative analysis examined the Herring Gull and Common Murre cases presented in Appendix A.

4 Results

Adaptation changed the form and length of the generated reports and was accompanied by higher automated ratings. We first report training behaviour and structural outcomes, followed by the rubric comparison and species-level observations.

4.1 Training Behaviour

Training loss decreased from 1.437 at optimisation step 10 to 0.376 at step 30. This decline showed that the training objective decreased during the run. Held-out structural results provide the separate assessment of whether the resulting adapter followed the reporting format on species outside the training subset.

4.2 Held-Out Structural Evaluation

All 15 adapted responses were valid JSON and contained the required keys, compared with none of the unadapted responses (Table 3). The base model returned Markdown prose under an instruction requesting a structured report, without explicitly requiring JSON. Parse failure and required-key failure therefore coincided. This weak baseline does not establish that adaptation is necessary for JSON output.

Table 3. Paired held-out results on 15 species. Rubric scores are automated five-point ratings.

Metric	Unadapted	QLoRA-adapted	Difference
JSON validity	0/15 (0.0%)	15/15 (100.0%)	+100.0 pp
Required-key presence	0/15 (0.0%)	15/15 (100.0%)	+100.0 pp
Mean output length	590 words	335 words	-255 words
Judge: format compliance	1.00	5.00	+4.00
Judge: factual grounding	2.33	5.00	+2.67
Judge: completeness	3.87	5.00	+1.13
Judge: reviewer usefulness	1.87	4.73	+2.86

The 95% exact binomial intervals were 0.0-21.8% for the unadapted proportion and 78.2-100.0% for the adapted proportion. Thus, although the observed outcomes were uniform within each condition, the small held-out set left substantial uncertainty about performance on additional species.

Adapted reports were shorter, averaging 335 words compared with 590 words for the unadapted responses. The judge also assigned higher mean scores to adapted reports across the four rubric dimensions (Figure 3). Grounding increased from 2.33 to 5.00, completeness from 3.87 to 5.00, and reviewer usefulness from 1.87 to 4.73. Format compliance increased from 1.00 to 5.00. These were descriptive automated ratings; variation in the individual scores was not analysed.

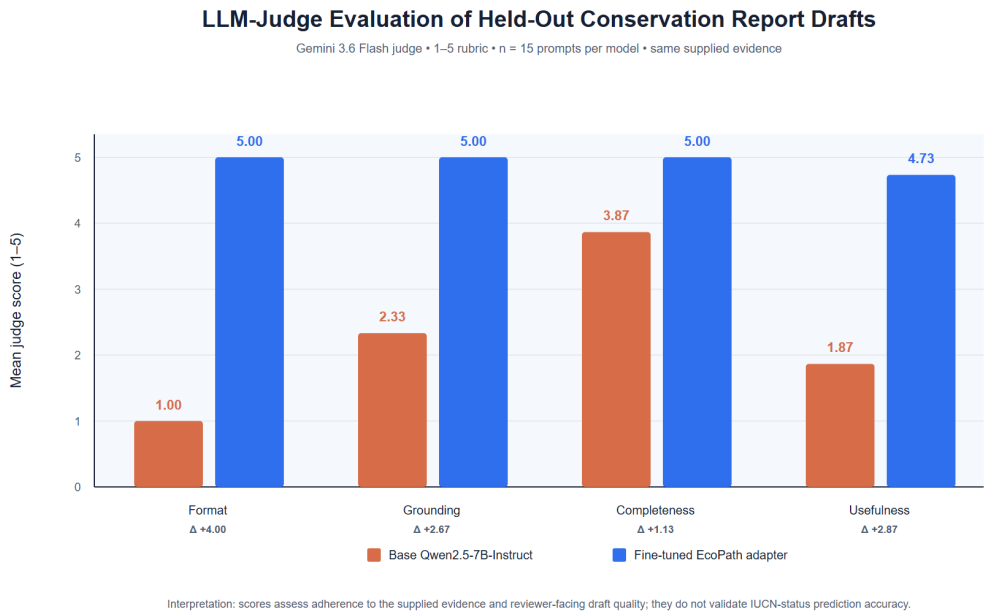


Figure 3. Historical automated ratings. Judge identity is unverified; context was limited to 600 prompt characters, generally omitting source excerpts. Scores do not validate full-evidence fidelity.

4.3 Per-Species Results

The structural difference was consistent across the 15 held-out species: every unadapted response failed the checks, and every adapted response passed. Adapted reports were also shorter for each species. The species list and two illustrative report comparisons are provided in Appendix A.

5 Discussion

5.1 Interpreting the Structural and Rubric Gap

The archived comparison shows that a small synthetic training set could establish a learned JSON reporting pattern under the tested instruction. A structured response can be parsed and presented through a consistent interface. However, the instruction did not explicitly require JSON, so the result does not establish superiority over strong prompting. The matched comparison in Appendix B evaluates the adapter under the same explicit prompt and output constraints as its unadapted comparator.

The supplementary ratings favoured the adapted reports, but their interpretation is limited by the judge context: only a 600-character prompt prefix was supplied, generally excluding source excerpts. These scores cannot show that the judge checked each report against the complete supplied evidence. Structure, brevity and similarity to the target style may also have influenced the ratings. We retain them as historical descriptive observations and use the direct structured-source checks in Appendix B to examine evidence agreement without treating automated judging as ecological validation.

The distinction becomes particularly important in this dataset. When the same ecological fields are missing throughout, a model can learn to mark them unavailable without learning when to report a supported value. What appears to be appropriate restraint under sparse inputs could become excessive abstention under richer ones. This is the central unresolved question raised by the findings: did the adapter learn a useful evidence-sensitive behaviour, a consistent template, or a combination of the two?

5.2 Qualitative Analysis

The Herring Gull (*Larus argentatus*) and Common Murre (*Uria aalge*) cases illustrate why the distinction matters. The unadapted reports introduced threats such as deforestation and wetland drainage without corresponding support in the input. The adapted reports instead identified missing information and retained the GBIF source reference. In the Common Murre case, the adapted response also explained that the available material could not support an IUCN-style assessment.

For a reviewer, the practical difference is between a plausible account that requires verification and an explicit account of what still needs to be established. The adapted examples better expressed the latter role. Nevertheless, two illustrative cases cannot show how often the same behaviour occurred across all factual statements. Source retention also establishes traceability only at the report level; it does not demonstrate that each claim is supported. Claim-level analysis is needed to connect these examples to an estimated error rate.

5.3 Relationship to Prior Work

Our findings fit the task-specific adaptation enabled by LoRA and QLoRA [4,6]. The contribution here is an application in which report structure and missing-information statements can be learned from a small dataset. Castro et al. [9] demonstrate the promise of prompt-based ecological processing, motivating a strong prompting comparator. Appendix B supplies this control while keeping the evidence and schema constraints identical between the unadapted and adapted conditions.

The ecological benchmarks of Dorm et al. [12] and Uryu [13] place the result in a broader context. They evaluate knowledge and judgement tasks that differ from drafting a report from supplied metadata. EcoPath did not predict verified conservation categories or infer population trajectories, and its structural results should not be compared numerically with those ecological prediction scores. The relevant connection is the need to define the task narrowly enough that successful performance has a clear meaning.

Tam et al. [10] motivate distinguishing format adherence from task success, while Zheng et al. [8] explain why automated judgements require scrutiny. The follow-up evaluation in Appendix B holds the output contract constant and compares the adapted model with a strongly prompted unadapted model on available and missing evidence. Its structured-source agreement measures address the original experiment's evidence-use gap, while remaining distinct from ecological or practical validation.

5.4 Role within the Wider EcoPath Platform

EcoPath connects information exploration with report preparation. Search, species profiles, and maps help users inspect available records; the drafting module offers a way to organise selected inputs for subsequent review. The structural improvement observed here supports that interface role. The study did not evaluate whether the reports reduced reviewer effort, improved decisions, or increased trust appropriately, and these outcomes remain separate application questions.

The most defensible use is therefore an initial draft that makes missing evidence visible. A report should help a reviewer identify the information available and the questions that remain, with suggested data-collection actions distinguished from species facts. Formal conservation interpretation continues to require appropriate evidence and expertise.

5.5 Limitations

The small, synthetic dataset limits the range of conclusions. Training and evaluation examples came from the same generation pipeline, and the split did not establish generalisation across taxonomic groups. Species-selection criteria were not recorded sufficiently to characterise sampling bias. Although the platform was motivated by Indian bird information needs, the held-out species did not constitute an India-specific evaluation. The findings therefore concern the sampled reporting examples rather than a regional conservation population.

The original evidence design limited the interpretation of missing-information handling: verified status, trend, population, habitat and range values were absent, so selective abstention could not be distinguished from a repeated unavailable-field response. Appendix B varies availability for six collected fields and measures supplied-value agreement and abstention jointly; wider ecological validity remains untested.

The archived comparison included one weak prompting condition, one training run and no independent validation set. Its incomplete runtime and hardware record limits reproducibility and cost analysis. Appendix B adds a pinned model revision, validation-based checkpoint selection and a matched constrained baseline. Only seed 42 was evaluated there; although three adapters were trained, the deadline-reduced comparison cannot estimate variation across training seeds.

Historical judge scores were not validated against blinded human review. The 600-character context limit, unverified judge identifier and unresolved teacher-judge independence prevent interpreting the rubric as calibrated ecological correctness. The newer checks compare output values and citations with reviewed structured assertions; source review was AI-assisted, not independent ornithologist validation. Neither experiment establishes reviewer benefit or formal conservation-assessment accuracy.

6 Conclusion

We studied QLoRA adaptation for structured bird report drafting in EcoPath. Archived 15-species adapted outputs produced valid JSON and required keys, whereas the base model returned prose. A matched 60-species comparison under identical prompts and schema constraints evaluated evidence fidelity (Appendix B). The evaluated adapter improved automatic field success under these conditions; this does not establish ecological truth, reviewer benefit, or seed invariance.

Learning a reporting pattern and using supplied evidence remain separate outcomes. The follow-up comparison measures available-value agreement, abstention and structural failures directly, while its single evaluated seed and dated, selected dataset limit generalisation. Further training-seed replication and independent ecological assessment remain outside the measured scope.

For EcoPath, the immediate implication is a bounded drafting role. A useful report should organise what is known and make the remaining evidence needs easier to inspect. The present results support the feasibility of that workflow while leaving its broader factual reliability and practical value to be established.

Acknowledgments

The authors thank Dr. Indu Dokare, Department of Computer Engineering, Vivekanand Education Society's Institute of Technology, for guidance and supervision of the project.

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A Archived Held-Out Evaluation Materials

Table 4 lists the 15 held-out species and their structural outcomes. Table 5 summarises the Herring Gull and Common Murre comparisons used to examine missing-information handling. The cases illustrate report behaviour and do not constitute a systematic estimate of unsupported-claim frequency.

Table 4. Structural outcomes for the 15 held-out species.

Held-out species	Base: JSON / keys	Adapted: JSON / keys
Herring Gull	Fail / Fail	Pass / Pass
Mediterranean Gull	Fail / Fail	Pass / Pass
European Goldfinch	Fail / Fail	Pass / Pass
Eurasian Hobby	Fail / Fail	Pass / Pass
Eurasian Hoopoe	Fail / Fail	Pass / Pass
Common Murre	Fail / Fail	Pass / Pass
Southern Cassowary	Fail / Fail	Pass / Pass
Crested Lark	Fail / Fail	Pass / Pass
Tawny Owl	Fail / Fail	Pass / Pass
Song Thrush	Fail / Fail	Pass / Pass
Common House-Martin	Fail / Fail	Pass / Pass
Peregrine Falcon	Fail / Fail	Pass / Pass
European Turtle-Dove	Fail / Fail	Pass / Pass
Eurasian Collared-Dove	Fail / Fail	Pass / Pass
Barn Swallow	Fail / Fail	Pass / Pass

Table 5. Illustrative paired reports under sparse evidence.

Species and supplied evidence	Unadapted output	Adapted output
Herring Gull (*Larus argentatus*). GBIF-linked identity and source reference; key ecological fields absent.	Markdown prose. Introduced deforestation without input support. Automated grounding score: 2/5.	Valid JSON. Marked missing fields unavailable, retained the source, and recommended data collection. Automated grounding score: 5/5.
Common Murre (*Uria aalge*). Same sparse evidence boundary.	Markdown prose. Introduced wetland drainage and deforestation without input support. Automated grounding score: 1/5.	Valid JSON. Stated that an IUCN-style assessment could not be completed and requested more evidence. Automated grounding score: 5/5.

Cases were not randomly selected. Their historical judge scores use truncated prompt context and do not measure complete-evidence fidelity; these examples do not estimate factual error frequency.

B Matched Held-Out QLoRA Evaluation

B.1 Dataset and matched conditions

To address the same report-drafting purpose with informative ecological inputs, we curated a separate 120-species India-frame dataset from State of India's Birds 2023. Development, validation and test partitions contained 30, 30 and 60 species, respectively, with no shared genera. Six fields could be supplied: recorded IUCN category, India-specific population trend, habitat, range-size category, diet and migration. Population size, breeding and species-specific threats remained null. Complete, partial and minimal variants exposed six, three and zero of the six supported fields; "complete" therefore means complete for the collected evidence, not a comprehensive conservation assessment.

The new adapter used 90 development examples (30 species x three variants) and 90 validation examples. Targets copied supplied values and source IDs, using null otherwise. QLoRA used rank 16, alpha 32, dropout 0.05, the seven projection targets in Section 3.4, learning rate 2×10^{-4}, batch one, accumulation eight and three epochs (36 optimizer steps). Training used the native chat template, assistant-only loss, a compact schema representation and no sequence truncation. Epoch checkpoints were selected by validation loss. Three seeds were trained; seed 42, fixed as the first planned seed before either matched test arm was generated, alone was evaluated for this submission.

Both conditions used Qwen2.5-7B-Instruct revision a09a35458c702b33eeacc393d103063234e8bc28, a 4-bit NF4 backbone with double quantization and fp16 computation, identical full engineered prompts and one validation demonstration, and the same LM Format Enforcer schema constraint. Greedy decoding used a 4,096-token input budget and 768 new tokens, with no evidence truncation. Each condition attempted all 180 held-out variants in its own process on an 8 GiB RTX 3070 Laptop GPU.

B.2 Paired results

End-to-end field success was 100.0% for QLoRA and 72.1% for the unadapted comparator (paired difference +27.9 percentage points; 95% species bootstrap interval +27.2 to +29.0).

Table 6. Matched test results (60 species, 180 outputs per condition). Counts retain structural failures. Agreement refers to reviewed structured assertions.

Metric	Unadapted	QLoRA seed 42
Valid JSON	179/180 (99.4%)	180/180 (100.0%)
Full-schema compliance	179/180 (99.4%)	180/180 (100.0%)
Field success, end to end	1168/1620 (72.1%)	1620/1620 (100.0%)
Available-value agreement, end to end	94/540 (17.4%)	540/540 (100.0%)
Correct abstention, end to end	1074/1080 (99.4%)	1080/1080 (100.0%)
False abstention, assessable fields	0/537 (0.0%)	0/540 (0.0%)
Cited structured-source agreement	94/537 (17.5%)	540/540 (100.0%)
Generation-cap hits	1	0
Execution errors	0	0

B.3 Evidence availability and interpretation

Table 7. End-to-end field success by evidence variant; each variant includes all 60 test species and 540 fields per condition.

Evidence variant	Unadapted	QLoRA seed 42
Complete	240/540 (44.4%)	540/540 (100.0%)
Partial	388/540 (71.9%)	540/540 (100.0%)
Minimal	540/540 (100.0%)	540/540 (100.0%)

The evaluated adapter improved automatic field success under these conditions; this does not establish ecological truth, reviewer benefit, or seed invariance.

A successful field contains the supplied value with the expected source association, or the required null when evidence is absent. Structural failures count as failures for all fields. Minimal variants have no available facts, so success there measures null handling rather than supported extraction. We therefore report available-value agreement alongside the aggregate score. Citation identifiers can be valid even when an output value differs from the corresponding source assertion; automated matching is not a semantic assessment of unrestricted prose or independent ecological truth.

Mean per-example wall time was 25.5 seconds for the unadapted condition and 29.4 seconds for QLoRA. These single-GPU, batch-one measurements exclude model loading and warmup and do not establish deployment throughput or amortized training cost. The paired effect averages the three variants within species and then across species. Its percentile interval uses 2,000 paired species bootstrap draws with seed 42, retaining variants together; it does not estimate training-seed variation.

B.4 Scope, provenance and reproducibility

The original seven-method, three-seed test plan was reduced for the submission deadline before generating these two prioritized arms. Other methods and seeds were deferred, and partial original-prompt outputs were preserved but excluded from this paired analysis. The original frozen protocol and its generation settings were retained. This amendment is disclosed; the submission is not presented as a completed seven-method comparison. The archived 15-species experiment uses a different dataset, adapter and report contract and is not pooled with these results.

The selected sampling frame is not a probability sample. All source values are from the fixed 2023 snapshot and are not claims about current conservation categories. Source IDs, URLs, retrieval dates, excerpts and rights metadata are retained; unavailable fields remain null. Source extraction review was AI-assisted, without independent ornithologist validation. The study does not measure conservation decision quality, reviewer usefulness or field-assessment accuracy. Synthetic targets, historical automated judging and AI assistance in source auditing and manuscript preparation are disclosed.

Source: State of India's Birds 2023, archived assessment table, Zenodo record 11124590 (https://zenodo.org/records/11124590). The recorded IUCN category is a quotation of that snapshot, distinct from India-specific priority classifications.

Frozen protocol SHA-256: b0fc3242e40019a55578a50c2d10f8663d4e5f0eab32a9bdc448270aea9e7d79. The saved protocol, dataset hash, model and adapter provenance, raw outputs, errors, metric denominators, deadline amendment and paired analysis are retained in the local reproducibility bundle. Public availability is not claimed without a release URL.